

# Ian Nicholas Ortega

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## Education

### MARQUETTE UNIVERSITY

**Major:** B.S. Biomedical Engineering

**Minor:** Computer Engineering

**GPA:** 3.9

Accelerating Ingenuity in Markets (AIM) Program, FinTech

Milwaukee, WI

Graduation - 2026

## Professional Experience

### RTX

*Digital Leadership Development Program Associate*

Arlington, VA

June 2026 – Present

- Support cross-functional **Pratt & Whitney** teams through development of **artificial intelligence, software tools, cloud technologies, data analytics, and digital systems**.
- Participate in structured leadership and technical development, mentorship, and initiatives supporting RTX business objectives.

### Collins Aerospace

*AI Engineering Intern*

Cedar Rapids, IA

May 2025 – August 2025

- Developed an **LLM-powered document translation platform** using local AI models and **optical character recognition**, supporting approximately **\$350K in projected business value**.
- Built scalable **AWS and Databricks data pipelines** for distributed data analytics, accelerating engineering time-to-insight.
- Applied **Agile software development** practices to iteratively deliver on-premises translation and document-processing capabilities.
- Authored **AR/VR business use cases** supporting Collins Aerospace Digital Technology strategy and planning.

### GE HealthCare

*Edison Hardware Engineering Intern*

Waukesha, WI

May 2024 – August 2024

- Designed and manufactured a **CT blower test fixture** using Creo, 80/20 components, and machining equipment, improving testing repeatability and efficiency.
- Designed inclined-impact testing equipment in **SolidWorks** and authored technical SOP's for consistent engineering validation.
- Automated materials-testing data processing and analysis in **MATLAB**, reducing manual effort and improving data accuracy.

### Computational Prototyping and Research Center

*CAD Modeler & Simulation Analyst*

Aurora, IL

January 2021 – May 2022

- Performed **finite element analysis** and **computational fluid dynamics** simulations to evaluate aerospace and hypersonic designs.
- Developed detailed CAD models and technical drawings based on sponsor requirements and engineering specifications.
- Validated design functionality through simulation and recommended design iterations during technical reviews.

## Research Experience

### Accelerating Ingenuity in Markets Program

*AI Research Assistant*

Milwaukee, WI

August 2024 – Present

- Co-authored and presented research on a **hybrid Retrieval-Augmented Generation (RAG)** system for NASA technical literature, integrating **dense embeddings, BM25 sparse retrieval, and Reciprocal Rank Fusion** to improve scientific document retrieval.
- Co-authored a follow-on study evaluating **Hypothetical Document Embeddings (HyDE)** for query enhancement, benchmarking retrieval performance across **250 technical queries**.

## Leadership and Projects

### Engineers Without Borders

*Project Lead*

Milwaukee, WI

September 2022 – May 2026

- Led CAD and structural design for a **\$350K clean-water infrastructure project** delivering across two communities in Guatemala.
- Awarded the 2026 NCEES Engineering Education Award and **\$25k grand prize** for a community-focused engineering solution.
- Supported the design of spring-capture, storage, and distribution infrastructure serving approximately **468 household taps**.

### Water Quality Prediction System

**Rapid Radicals**

Milwaukee, WI

August 2025 – May 2026

- Developed a preliminary machine-learning workflow sensor data to predict *E. coli* contamination.

Milwaukee, WI

March 2025 – July 2025

- Modeled a novel water treatment system as a private consulting contractor for Rapid Radicals.

**Technical:** CSWA (Certified SolidWorks Associate), C, Java, React, Excel VBA, MATLAB, SQL, Python

**Interests:** Machine Learning, AI Integrations, Mechanical Design, Systems Engineering, Flow Simulations



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